Day	Backend Task	DSA Topic & Problems	System Design Topic	Output / Proof of Work
1	**Express setup, 3 routes, middleware, error handler**	**Arrays + Hashmap (Two Sum, Contains Duplicate)**	**Request → Server → DB flow**	**API responds, 2 LC problems done**
2	Controllers + services, folder structure	**Arrays (Buy/Sell stock, Majority Element)**	Latency, throughput, stateless servers	Modular API structure, 2 LC problems
3	Full CRUD (in-memory)	**Sliding Window (Longest substring)**	Cache vs DB	CRUD complete, 1 LC problem
4	Connect DB (Postgres/Mongo), convert CRUD	Sliding Window (Min window substring)	DB replication	DB CRUD working, 1 LC problem
5	Pagination, filtering, search	Stack/Queue (Valid parentheses, Min stack)	Indexing	Pagination working, 2 LC
6	JWT Auth + bcrypt + protect routes	Linked List (Reverse, Detect cycle)	JWT vs Session	Auth working, 2 LC
7	Deploy Notes API + README + Postman collection	Binary Search	CDN	Deployment live
8	File uploads (Cloudinary/S3)	Trees (DFS preorder/inorder/postorder)	Vertical vs Horizontal scaling	Uploads working
9	Redis caching (LRU)	Trees (LCA, depth, height)	Cache invalidation	Cache demo + 2 LC
10	Queue + background jobs	BST questions	Producer/Consumer	Queue job visible in logs
11	Logs + analytics middleware	Heap / Priority Queue	Log Aggregation	Logs tracked

12	Rate limiter middleware	Graph (BFS + DFS)	Token bucket	Rate limiter working
13	CI/CD pipeline (GitHub Actions)	Topo Sort + cycle detection	Monolith vs Microservices	Auto deploy setup
14	Social Feed backend base (posts + likes + comments)	Recursion	Feed high-level design	Repo + commit pushes
15	Refresh token + secure cookies	Tree DP	Consistent hashing	Refresh works
16	WebSockets notifications	Tree DP more	WebSocket scaling	“Notification received” demo
17	RBAC (admin routes)	Dijkstra shortest path	RBAC patterns	Admin routes working
18	Transactions & race conditions	DP (0/1 knapsack)	Read-write splitting	Transaction demo
19	Horizontal scaling + session store	DP subsequences	Load balancer algorithms	Session survives restart
20	Metrics + monitoring (Prometheus/Grafana)	DP on grids	Observability	Dashboard screenshot
21	Social Feed upgrade + WebSockets + RBAC + Monitoring	Revise weak areas	Review scaling tradeoffs	Final feed backend v1
22	URL shortener microservice	Sliding window + binary search	URL shortener deep dive	Microservice deployed
23	Rate limiter on feed backend	Graph revision	Rate limiting architecture	Rate limit live in prod
24	Chat backend (send/receive + seen)	Trees + graphs mixed	Chat scaling	Demo of chat

25	Table sharding / partitioning	DP set 2	Sharding strategies	Sharding notes + code
26	Feed ranking + caching	Mock set (4 problems)	News feed deep dive	Faster feed API
27	README + architecture diagrams + docs	3 contest problems	CAP theorem	Updated GitHub
28	Backend timed mock interview	Timed 2 LC	System design mock	Screen recording (optional)
29	Fix weak areas + deploy stable final	3 timed questions	Review scaling patterns	Final deployment
30	Apply to 20+ roles + interview prep	Re-solve 5 failed LC problems	Final revision	Job applications submitted